Hasan Zaki Rizvi

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EDUCATION

Curtin University | Bachelor of Advanced Computer Science Honours | 79 CWA 2024 - Current

- Coursework: Database Systems, Data Structures and Algorithms, Machine Learning, Operating Systems
- Letter of Commendation (2025)

Willetton Senior High School | 96 ATAR 2018 – 2023

- Coursework: English, Methods, Chemistry, Human Biology, Physics

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, SQL, C, HTML/CSS
- **Tools:** Flask, SQL/PostgreSQL, NumPy, Pandas, RAG, LangChain, FAISS, PyTorch, Git, NeoVim, Linux (Ubuntu), TensorFlow, Hugging Face, SQLite, Basiq API

PROJECTS

Smart Expense Tracker & Banking Analytics App June 2026 – Current

Python, FastAPI, PostgreSQL, Pandas, NumPy, Scikit-learn, OpenAI API, Streamlit, Basiq API

- Implementing a Python-based personal finance analytics platform that securely retrieves live banking transaction data through the Basiq Open Banking API and stores structured financial records in a PostgreSQL database
- Developing automated spending categorisation, subscription detection, savings-rate analysis, and anomaly detection using data-processing pipelines, rule-based logic, and machine learning techniques
- Building an interactive analytics dashboard with financial visualisations, transaction filtering, budgeting insights, and AI-generated monthly summaries using OpenAI models
- Implementing a modular FastAPI backend with SQLAlchemy, REST APIs, and scalable analytics components to support real-time financial reporting and future multi-bank integration

VALORANT Comms & Pattern Analysis AI Tool June 2026 – Current

Python, WhisperX, pyannote, Hugging Face, OpenAI API

- Building a local AI-powered VALORANT comms analysis tool that captures microphone, Discord, and in-game voice audio for continuous transcription and post-round review
- Engineering a speech-processing pipeline using WhisperX, faster-whisper, and pyannote diarization to separate comms by speaker and organise transcripts into structured round-based data
- Building an LLM-powered behavioural analysis pipeline using RAG, prompt engineering, and machine learning to infer tactical patterns and predict player movement and rotations from historical communication and gameplay data

GenAI for Controlled Human Mobility Data Generation Feb 2026 – Current

Python, FAISS, OpenAI API, RAG, NumPy, Pandas, Matplotlib

- Processing large-scale mobility data into structured trajectory segments and generating controlled synthetic afternoon/night trajectories using a RAG pipeline with FAISS retrieval, OpenAI embeddings, and prompt-based generation
- Implementing a RAG-based retrieval system using FAISS, OpenAI embeddings, and cosine similarity/DTW reranking to retrieve similar mobility patterns and guide trajectory generation
- Developing an evaluation workflow comparing generated trajectories against ground-truth mobility data using DTW, cosine similarity, endpoint distance, grid overlap, and visual trajectory analysis
- Conducting the research project under the direct supervision of Dr Chamara Kattadige (University Professor), following an iterative research and development process with regular technical reviews and project milestones

AI Virtual Health Assistant for CALD Healthcare Support Feb 2025 – Oct 2025

Python, Flask, LangChain, FAISS, OpenAI API, RAG

- Developed a cross-platform AI healthcare assistant using Flutter, Flask, and SQLite to support culturally and linguistically diverse users with accessible health information
- Built a LangChain and FAISS-based RAG system for document-grounded healthcare responses, improving answer reliability through prompt engineering, safety filters, and human review workflows
- Designed structured local data storage and retrieval workflows to manage user queries, healthcare documents, and AI-generated responses across the mobile application
- Completed the project under the direct supervision of Dr Susannah Soon (University Professor) contributing to the design, implementation, and evaluation of an applied AI research solution

WORK EXPERIENCE

Seafood Processor | KB Seafood Oct 2024 – Current

- Processed crayfish products to strict food safety and SOP standards while operating specialised machinery in a high-volume production environment
- Used WMS for inventory tracking and order fulfilment, maintaining high accuracy and meeting daily production and shipping targets

Volunteer Store Helper | Save the Children - Op Shop July 2025 – Current

- Assisted customers in-store, processed transactions and maintained high standards of service
- Supported store operations including stock receiving, pricing, replenishment, and housekeeping

Volunteer Helper | Local Community Events 2019 – 2021

- Operated audio systems for effective sound management & Helped with setup & cleanup

Leadership

Valorant Premier Team | Captain & Coach Sept 2023 – Nov 2025

- Developed team strategies to improve competitive performance, including analysing opponents, planning agent compositions, and coordinating utility usage during matches
- Coached teammates through targeted feedback, VOD reviews, and practice routines while improving communication, teamwork, and in-game decision-making

Research Project Team Leader | University Group Project Feb 2024 – Jun 2024

- Coordinated a team of students to deliver a 4,000-word research report over a semester-long project
- Assigned responsibilities, tracked progress, and ensured deadlines were met
- Integrated and reviewed team contributions to maintain consistency and quality across the final submission

HOBBIES & LANGUAGES

- Gaming, PC Building, Calisthenics, Gym, Hiking
- English (Native), Urdu (Native), French (Elementary), Arabic (Beginner)